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Towards transfinite type theory: rereading Tarski's *Wahrheitsbegriff*

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Abstract In his famous paper *Der Wahrheitsbegriff in den formalisierten Sprachen* (Polish edition: Nakładem/Prace Towarzystwa Naukowego Warszawskiego, wydzial, III, 1933), Alfred Tarski constructs a materially adequate and formally correct definition of the term "true sentence" for certain kinds of formalised languages. In the case of other formalised languages, he shows that such a construction is impossible but that the term "true sentence" can nevertheless be consistently postulated. In the Postscript that Tarski added to a later version of this paper (Studia Philosophica, 1, 1935), he does not explicitly include limits for the kinds of language for which such a construction is possible. This absence of such limits has been interpreted as an implied claim that such a definition of the term "true sentence" can be constructed for every language. This has far-reaching consequences, not least for the widely held belief that Tarski changed from an universalistic to an anti-universalistic standpoint. We will claim that the consequence of anti-universalism is unwarranted, given that it can be argued that the Postscript is not in conflict with the existence of limits outside of which a definition of "true sentence" cannot be constructed. Moreover, by a discussion of transfinite type theory, we will also be able to accommodate other of the changes made in Tarski's Postscript within a type-theoretical framework. The awareness of transfinite type theory afforded by this discussion will lead, in turn, to an account of Tarski's Postscript that shows a gradual change in his logical work, rather than any of the more radical transitions which the Postscript has been claimed to reflect.

Keywords Tarski · Truth · Type theory · Universalism

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1 Introduction

Even after 80 years, Alfred Tarski's well-known paper "*Pojęcie prawdy w językach nauk dedukcyjnych*" (Tarski 1933)[1] remains central to any adequate understanding of the development of logic in the first half of the twentieth century. In this paper, Tarski constructs a definition of "true sentence" for some formalised languages, while showing the impossibility of constructing such a definition for others.

The German version (Tarski 1935) and the English version based upon it (Tarski 1956)[2] both include an additional postscript (henceforth: *Ps*), which extends the results of the body of the text (henceforth: *Wb*) and hence of the Polish original.[3] A very significant change is that Tarski abandons the so-called theory of semantical categories in *Ps*, which in its turn leads to a modification of the conclusions of *Wb*.

If we compare the conclusions of *Wb* and *Ps*, one difference is immediately apparent: whereas *Wb* contains three conclusions, *Ps* contains only two. One of the conclusions of *Wb* is that for certain languages a construction of the definition of *true sentence* is simply not possible. Such an undefinability conclusion is not to be found in *Ps*.

This absence is explained in de Rouilhan (1998), according to whom Tarski believed that one could construct a definition of *true sentence* for every formalised language. This would have a number of deep consequences. Because the definition of *true sentence* should be given in a language of a higher order than that of the language under consideration, it follows that there is, so to speak, no language of the highest order: there would not be a universal language. Consequently, because a Russellian type-theoretic framework goes together naturally with a universalistic point of view, it follows that Tarski had moved away—at least in spirit—from type theory in *Ps* (assuming that Tarski was working in a type-theoretical framework in *Wb* in the first place). De Rouilhan gives some additional reasons to suggest that Tarski had also moved away syntactically from type theory.

These are profound consequences indeed: not because of a possible change in the framework of logic in itself, but because of the (radical) change of philosophical position that seems to go with it. To conclude that such a transition has taken place accordingly calls for solid argumentation.

In the current paper we will argue that the languages considered by Tarski in *Ps* may well be type theoretical (see also Gómez-Torrente 2001, pp. 161–162; Rodríguez-Consuegra 2005, p. 231), possibly using transfinite types.

Our argumentation consists of two parts. First we will use Field's idea (see Field 2008) to counter de Rouilhan's argument that there is no universal language in *Ps*. The purpose of the current paper is not to defend Field's idea, but this idea is used rather as a reasonable departure for our argumentation that the transition to anti-universalism/set

[1] German version: "*Der Wahrheitsbegriff in den formalisierten Sprachen*" Tarski (1935); English edition: *"The Concept of Truth in Formalized Languages"* Tarski (1956).

[2] For a comparison between the Polish, German and English versions see Gruber (2012).

[3] It may be argued that the choice of the abbreviations "*Wb*" and "*Ps*" is not very consistent, language-wise. Yet we will keep to these in order to facilitate the discussion of de Rouilhan (1998), who uses the same abbreviations.

theory cannot be read from *Ps* without doubt. This first part shows that the claim of anti-universalism is unwarranted.

In the second part of our argumentation we will show that the syntactical properties of the languages in *Ps* can also be explained within a type theoretical framework. Specifically we will argue that expressions of transfinite order and variables of indefinite order naturally occur in transfinite type theory. Thus also the claim that Tarski had moved to set theory in *Ps* is not watertight.

This type-theoretical account of *Ps* refutes not only the alleged transition from type theory (*Wb*) to set theory (*Ps*) (see also Sundholm 2003, pp. 119–120), but also puts the claimed transition from type theory (*Wb*) to Carnap's theory of levels (*Ps*) (Patterson 2012, p. 192) in a broader perspective. Although, as we shall see, certain ideas from Carnap's theory of levels also play a role in our type theoretical account of *Ps*, these ideas are not unique to Carnap's theory of levels and thus cannot justify the claim of a complete transition from the one framework to the other. Instead, the type theoretical account witnesses a gradual change, we argue, from a Leśniewskian framework already pervaded by ideas from Russell's type theory (*Wb*) (see Betti 2008, p. 55) to transfinite type theory (*Ps*).

2 Wahrheitsbegriff

The aim of *Wb* is to construct a materially adequate and formally correct definition of "true sentence" (Tarski 1956, p. 152), without the use of semantical concepts that cannot be reduced to other concepts (Tarski 1956, p. 153). Tarski focuses in most of the paper on formalised languages and distinguishes between object language and metalanguage.

All formalised languages fulfill the same general syntactical requirements for Tarski in *Wb*, though for our current purpose it is most important that the metalanguage does. The most essential syntactical requirement for our current purpose is Tarski's use of the theory of *semantical categories*, which originates directly from their use in Leśniewski's systems (Leśniewski 1929). Tarski writes:

> This concept [of *semantical category*], which we owe to E. Husserl, was introduced into investigations on the foundations of deductive sciences by Leśniewski. From the formal point of view this concept plays a part in the construction of a science which is analogous to that played by the notion of type in the system *Principia Mathematica* of Whitehead and Russell. But, so far as its origin and content are concerned, it corresponds (approximately) rather to the well-known concept of part of speech from the grammar of colloquial language. Whilst the theory of types was thought of chiefly as a kind of prophylactic to guard the deductive sciences against possible antinomies, the theory of semantical categories penetrates so deeply into our fundamental intuitions regarding the meaningfulness of expressions, that it is scarcely impossible to imagine a scientific language in which the sentences have a clear intuitive meaning but the structure of which cannot be brought into harmony with the above theory. (Tarski 1956, p. 215)

It is important to note what Tarski does *not* claim here about the relation between semantical categories and Russell's types. First, he does not claim that they amount to the same thing. This is correct, as there exists a very important difference between the two: the former applies to *expressions*, whereas the latter applies to *objects*,[4] in a manner somewhat more similar to the ontological ordering in set-theory that we will meet in Sect. 4. Second, he does not claim that in *Wb* he is working within a Russellian type theory; neither is it implied in the footnote (Tarski 1956, fn 1, p. 215), which states that the theory of semantical categories is less remote to the simple theory of types than to the ramified theory of types in *Principia Mathematica*.[5] Third, and most importantly, Tarski does not claim that abolishing the principles of the theory of semantical categories as he does in *Ps* (which we will discuss below) will reduce the similarities between the formal systems under consideration and Russellian type theories. On the contrary, it seems to me that the principles of semantical categories are essential for *Leśniewski's systems*, and by taking a more liberal stance towards these categories it may even be argued that the gap between these formal systems and the theory of types is reduced (see Sect. 5) instead of being widened as others have argued (see Sect. 4).

Tarski doesn't explain which expressions belong to which semantical categories; instead he explains what is the case when two expressions *belong to the same semantical category*. They do if (Tarski 1956, p. 216):

1. There is a sentential function[6] which contains one of these expressions;
2. No sentential function which contains one of these expressions ceases to be a sentential function if this expression is replaced in it by the other.

Based on the above requirement, in order to conclude that two expressions belong to the same category one should check *all* sentential functions in which they occur for the admissibility of swapping the two expressions. According to the *First principle of the theory of semantical categories*, which Tarski subsequently assumes, it suffices to check only one:

> In order that two expressions belong to the same semantical category, it suffices if there exists *one* [sentential] function which contains one of these expressions and which remains a function when this expression is replaced by the other. (Tarski 1956, p. 216)

Also for primitive sentence forming functors[7] there is a law, the *Law concerning the semantical categories of sentence-forming functors*, by which one can more easily establish whether they belong to the same category:

[4] Although by *abuse de langage* Tarski also applies the notion of semantical categories to objects (Tarski 1956, p. 219), and although the situation in Russell is hard to assess unambiguously.

[5] Compare Feferman (2008, p. 85) and Sundholm (2003) (only on p. 119), who seem to assume that the theory of *Wb* is or "contains" the simple theory of types.

[6] A sentential function is an open or closed formula. This notion is used to differentiate those expressions from, for example, terms of the language.

[7] Functors are signs denoting functions.

> The functors of two primitive sentential functions belong to the same category if and only if the number of arguments in the two functions is the same, and if any two arguments which occupy corresponding places in the two functions also belong to the same category (Tarski 1956, p. 217).

These two principles are assumed in *Wb*, but abolished in *Ps*

Next Tarski defines the *order of a semantical category*, a classification of the semantical categories as follows (Tarski 1956, p. 218):

- Individuals and variables representing them are of the first order;
- Functors of primitive functions of which all arguments are at most of the nth order and at least one of them is exactly of the nth order, are of order $n+1$.

All expressions which belong to a certain semantical category are also said to have its order.

Together, semantical categories and the orders of expressions form the basis to distinguish four kinds of languages, and Tarski discusses the construction of the definition of a true sentence in relation to each of these four, one at a time (Tarski 1956, p. 220)[8]:

1. Languages in which all the variables belong to one and the same semantical category;
2. Languages in which the number of categories in which the variables are included is greater than 1 but finite;
3. Languages in which the variables belong to infinitely many different categories, but in which the order of these variables do not exceed a previously given natural number n;
4. Languages which contain variables of arbitrarily high order.

Using this categorisation of languages, Tarski's summarises his results as follows (Tarski 1956, pp. 265–266):

A *For every formalized language of finite order a formally correct and materially adequate definition of true sentence can be constructed in the metalanguage, making use only of expressions of a general logical kind, expressions of the language itself as well as terms belonging to the morphology of language, i.e. names of linguistic expressions and of the structural relations existing between them.*

B *For formalised languages of infinite order the construction of such a definition is impossible.*

C *On the other hand, even with respect to formalized languages of infinite order, the consistent and correct use of the concept of truth is rendered possible by including this concept in the system of primitive concepts of the metalanguage and determining its fundamental properties by means of the axiomatic method.*

Especially conclusions *B* and *C* are relevant to us, as we will see that analogous conclusions are absent in *Ps*. It is from this absence that various incompatible interpretations of *Ps* have originated.

[8] Tarski also defines the notion of *semantical type* (Tarski 1956, p. 219), which is based on the number of free variables and the semantical categories of those variables given some sentential function. Tarski's notion of semantical types is thus very different from the notion of type in Russell's type theories.

3 Tarski's Postscript

The fact that no definition of *true sentence* could be constructed for languages of infinite order may be blamed—and Tarski did blame it—on the lack of expressions of infinite order. If we could solve this latter problem, then, we may also be able to construct a definition of *true sentence* for these.

In the postscript that Tarski first added to the German version (Tarski 1935), he seems, as many have also argued elsewhere, to have found a way to solve the problem of the lack of expressions of infinite order. This view is expressed, for example, in Frost-Arnold (2004, p. 272), Feferman (2008, p. 86) and Creath (1999, p. 96), but is in disagreement with the view expressed in Patterson (2012, p. 191).

Let me sketch briefly an account of Tarski's solution in *Ps* and of its consequences for his notion of formalised languages. In *Ps* Tarski had figured out a way to deal with infinity and thus to generalise the notion of order to infinite ordinals,[9] but it appeared to him then that this was not in itself enough to construct languages which actually were of order α, where α is some transfinite ordinal (i.e. an infinite ordinal number). Were variables of indefinite order not also allowed, one wouldn't obtain "languages which are actually superior to the previous languages in the abundance of the concepts which are expressible by their means, and the study of which could throw new light on the problems in which we are interested" (Tarski 1956, pp. 270–271). This can be understood in the following way: The order of a language is the least (ordinal) number higher than the order of any expression contained in the language. However, "the only way to get expressions of transfinite order is to have expressions that take arguments of *all* finite orders and hence to allow variability in the order of the arguments that a functional expression takes" (Patterson 2012, p. 192).

Allowing variables of indefinite order would not be possible, however, in a language which holds on to the theory of semantical categories. So Tarski let go of the principles of this latter theory. Having done away with these principles, Tarski proposes another method to allocate orders to expressions within formalised languages, and, ultimately, to these languages themselves. Because the principles of the theory of semantical categories are no longer assumed to hold, one and the same functor may in various occurrences hold variables of various orders in the same places, making it impossible to allocate orders to these functors based on the fixed order of the arguments as before. Instead, the order of such a functor is defined as the smallest number which is greater than the orders of all arguments in all sentential functions in which the functor occurs as a sentence-forming functor (Tarski 1956, pp. 269–270). The order of a language is then the smallest order which exceeds the order of all variables occurring in that language (Tarski 1956, p. 270). This solves the problem of allocating orders to signs in the absence of the principles of the theory of semantical categories. In order to allow expressions of infinite order in a language, one need only realise that "number" in the definition of order might as well be an infinite ordinal.

Now Tarski is able to draw the following conclusions (Tarski 1956, p. 273):

[9] Tarski refers to Fraenkel (1928). According to Carnap (1934, p. 189) the first to point out the possibility of transfinite 'levels' were Hilbert (1926, p. 184), and Gödel (1931, p. 191).

A* *For every formalized language a formally correct and materially adequate definition of true sentence can be constructed in the metalanguage with the help only of general logical expressions, of expressions of the language itself, and of terms from the morphology of language—but under the condition that the metalanguage possesses a higher order than the language which is the object of investigation.*

B* *If the order of the metalanguage is at most equal to that of the language itself, such a definition cannot be constructed.*

Note that Tarski has gone from three conclusions to two, but that only conclusion A^* can be seen as analogous to one of the conclusions of *Wb*. Conclusion B^* is not completely analogous to conclusion B of *Wb*, because the former does not state that for certain languages the construction of such a definition is impossible.

What can be derived from the conclusions that are actually there? If we connect statement A^* with the conviction expressed by Tarski that for any given object language it is possible to construct a meta-language which contains variables of an order higher than all variables in the object language (Tarski 1956, pp. 271–272), it seems valid to conclude that for *every* formalised language we can construct a definition of *true sentence*, full stop. It is here, however, that we should consider the question of the limits of the languages that Tarski was considering here. Field gives another, very relevant interpretation of the meaning of the conclusions of *Ps*:

> [I]f there is an ordinal bound on the orders of the variables in the language, truth is definable (in a language where there is no such bound or where the bound is higher). It is natural to make the further generalization that for those languages for which there isn't even an ordinal bound, truth is undefinable, and we must settle for an axiomatic theory of truth (in the form of an inductive definition). (Field (2008), pp. 35–36)

Field thus argues that, similarly to *Wb*, where we could construct the desired definition only for languages the order of which was limited by a finite ordinal, in *Ps* the languages for which such a definition can be constructed are restricted to those the order of which is limited by *any* ordinal. One would then expect the following conclusions $\overline{B}$ and $\overline{C}$, which are analogous to conclusions B and C of *Wb* respectively:

$\overline{B}$ *For formalised languages of which the order of the variables is unbound by any ordinal number, the construction of a formally correct and materially adequate definition of true sentence is impossible.*

$\overline{C}$ *On the other hand, even with respect to formalized languages of which the order of the variables is unbound, the consistent and correct use of the concept of truth is rendered possible by including this concept in the system of primitive concepts of the metalanguage and determining its fundamental properties by means of the axiomatic method.*

But these conclusions are absent from *Ps*. Instead, according to Field, *Ps* expends not a single word on the construction of the definition of *true sentence* for languages unbounded by any ordinal number: neither it is claimed that for such languages the definition can be constructed, nor that it cannot.

Why would Tarski be silent concerning this kind of languages? We cannot be sure. What we cannot exclude, however, is that a language which contains variables of any ordinal order[10] plays in *Ps* a role similar to that played in *Wb* by a language which contains variables of any *finite* ordinal number; that is, the role of *universal language*. We take "universal language" here to mean a language that contains expressions of every order.

Not only can we ask why perhaps Tarski didn't consider a certain kind of language; we can ask with even greater urgency what languages Tarski does in fact consider. Are there examples of languages for which the principles of the theory of semantical categories does not hold and of which the order is infinite? Tarski doesn't give us any. The language he does mention, the language of set theory, appears only in a footnote:

> From the languages just considered it is but a small step to languages of another kind which constitutes a much more convenient and actually much more frequently applied apparatus for the development of logic and mathematics. In these new languages all the variables are of indefinite order. From the formal point of view these are languages of a very simple structure; according to the terminology laid down in §4 they must be counted among the languages of the first kind, since all their variables belong to one and the same semantical category. Nevertheless, as is shown by the investigations of E. Zermelo and his successors (cf. Skolem 1929, pp. 1–12), with a suitable choice of axioms it is possible to construct the theory of sets and the whole of classical mathematics on the basis provided by this language. (Tarski 1956, fn 1, p. 271)

Note that Tarski doesn't identify the language of set theory as an example of the languages in the body of the text, though he does say it is "but a small step" from the latter to the former. This step includes changing to yet another notion of order, of which we will speak in the next section. The following statement is thus in my opinion untenably strong:

> In the *Nachwort* [*Postscript*], Tarski abjures this theory [(simple) type theory] and converts to set theory. (Sundholm 2003, pp. 119–120)

4 The abandonment of type theory?

As we have seen in the previous section, Tarski introduces in the Postscript a class of languages for which the fundamental principles of the theory of semantical categories no longer hold. Essential for the considered languages is the possibility of containing variables that "run through" all possible orders. We have also seen that Tarski offered no examples of such a language, but only some remarks in a footnote that it is a small step from these kinds of languages to the language of set theory.

Consequently some[11] have interpreted *Ps* as reflecting a transition in Tarski's thinking from type theory (in *Wb*) to set theory (in *Ps*). Moreover, it is not just the footnote on the languages of set theory that seems to suggest an abandonment of type theory, but rather a multitude of larger and smaller issues clearly and rather convincingly dis-

[10] It is, of course, quite hard to imagine what such a language would look like. See also the remark on the idealisation of languages in Linnebo and Rayo (2012, p. 277).

[11] E.g. Sundholm (2003) as quoted above.

cussed in de Rouilhan (1998). Nevertheless by discussing de Rouilhan (1998) below, I will argue against the view that Tarski was no longer working in type theory in *Ps*.

Even more far-reaching than the alleged abandonment of type theory is the idea, also put forward in (de Rouilhan 1998, p. 88), that *Ps* marks the beginning of a new era in logic, in which the approach is *model theoretical*, rather than *universal*. The distinction between these two traditions has been put forward by Hintikka (1988), and is a generalisation of Van Heijneoort's distinction between *logic as a language* and *logic as a calculus* (Heijenoort 1967). Yet exactly what it means to work according to the universal approach is not completely clear. (For several different uses of the notion of universality see Proops 2007 and Rodríguez-Consuegra 2005). It is thus important to be explicit on the precise meaning of universality in the current context.

In the context of Tarski's work, universality is often studied as manifested in a fixed approach of model domain, in contrast to a variable approach (for an overview see Mancosu 2010). Although related to the fixed vs variable domain issue, de Rouilhan speaks of the universalistic standpoint in a slightly different way. He understands it as the belief in the existence of a universal, formalised language (de Rouilhan 1998, p. 102). For the definition of a universal language de Rouilhan refers to Tarski's examples in *Wb*, from which the following similar descriptions of a universal language may be distilled[12]:

1. A universal, formalised language is a language of which all other formalised languages are either fragments, or can be obtained from it or from its fragments by adding certain constants, provided that the semantical categories of these constants are already presented by certain expressions of the given language (p. 210);
2. A universal language is a language of a complete system of logic: a language that contains—actually or potentially—all possible semantical categories which occur in the languages of the deductive sciences (p. 220).

Tarski is quite explicit in *Wb* concerning his belief in the existence of these languages. Note that both of these descriptions suppose that it is meaningful to speak of semantical categories in the considered language. In order to address the question of whether the languages of *Ps*, which might be set theoretical, are approached from a universalistic standpoint, a notion of order should be defined also for the language of set theory. This notion of order can then replace the notion of semantical category used in the above description of universal languages.

In de Rouilhan (1998) we find a detailed comparison between the languages that Tarski studied in *Wb*, the description of the new languages in *Ps*, and the language of set theory. The key observation is that whereas the notion of order that has been introduced for the former two cases is syntactical, a notion of an ontological order seems more fruitful for set theory. This idea is concisely explained already in (Tarski, 1956, fn 1, p. 271).

[12] Note that Rodríguez-Consuegra distills a different, but related, notion of universality from the same passages of *Wb* (Rodríguez-Consuegra 2005, p. 229). What he calls *mathematical universality* concerns the capacity of expressing the whole of mathematics, whereas de Rouilhan's notion revolves concerns the incorporation in a language of elements of all categories/orders. Moreover, Rodríguez-Consuegra considers Tarski's formalisation of set theory to be universal, because it additionally uses a fixed, single universe of discourse, instead of one for every type (p. 230/232). Because in this latter understanding of the notion of universality one could even say that type theory is *not* universal, it already shows that it is quite far from our use of universality in the current paper.

To see that the notion of syntactic order may not be very significant for set theory, it is enough to recall that it is in this sense a first-order language (Tarski 1956, fn 1, p. 271; de Rouilhan 1998, p. 90). This also means necessarily that all variables are of the first order. However, there is another notion of order—ontological order—in which variables, and in fact *all* variables of set theory, can be considered as variables of indefinite order, in total agreement with Tarski's footnote 1 cited above from *Ps*.

In Tarski's own words, in the second half of his footnote, we find the following description of ontological order:

> For the languages here discussed the concept of order by no means loses its importance; it no longer applies, however, to the expressions of language, but either to the objects denoted by them or to the language as a whole. Individuals, i.e. objects which are not sets, we call objects of order 0; the order of an arbitrary set is the smallest ordinal number which is greater than the orders of all elements of the sets whose existence follows from the axioms adopted in the language. (Tarski 1956, fn 1, p. 271)

De Rouilhan argues as follows for the claim that *Ps* evinces a change in attitude toward logic itself (p. 88), apparent in Tarski's exchange of the universalistic, type theoretical framework of *Wb* for an anti-universalistic, set theoretical one. He does not argue that the languages Tarski speaks of in *Ps* are those of set theoretical theories, nor that the *Ps* makes full sense if one interprets the languages that Tarski considers to be set theoretical. He argues merely, rather, that some statements in *Ps* make sense in a set theoretical setting, and do not make sense in a type theoretical one. It is this claim that I will challenge.

Although it seems that the change perceived by de Rouilhan is twofold—both in the universalistic point of view and the kind of languages—he argues that these are, in fact, closely interlinked:

> The framework of the simple theory of types had something rigid, which the one of set theory does not have. The idea of a superior framework to the theory of simple types, which we could, however, still qualify as "simple type theory" could not even come to mind: If one would change the framework, it would be radically changed. Now, however, it seems that one can always change the framework while preserving the set-theoretical character. The putative universality of a set-theoretical framework always seems able to be replaced by that of another of the same kind, and universalism loses all verisimilitude.[13] (de Rouilhan 1998, p. 101/102)

In other words, anti-universalism is implausible in the framework of the simple theory of types, whereas a set-theoretical framework may lead to an anti-universalistic point of view. Yet the abandonment of type theory (or semantical categories) is not enough

[13] Le cadre de la théorie des types simple avait quelque chose de rigide que celui de la théorie des ensembles n'a pas. L'idée d'un cadre supérieur à celui de la théorie des types simples, mais que l'on pût encore qualifier de "théorie des types simples" ne pouvait même venir à l'esprit: si l'on changeait de cadre, on en changeait radicalement. Maintenant, au contraire, il semble que l'on puisse toujours changer le cadre tout en préservant le caractère esembliste. L'universalité putative d'un cadre esembliste semble pouvoir toujours être supplantée par celle d'un autre de même genre, et l'universalisme perdre toute vraisemblance.

to justify a move away from universalism, de Rouilhan argues (p. 101); that move is, according to him, not mathematically grounded (p. 102).

De Rouilhan argues that the assumption that Tarski still had in mind a universalist, type-theoretical idea in *Ps*, does not hold (by comparing it to an imaginary version of *Ps* in which a certain version of set theory plays the role of universal language). This can be analysed in two steps: first by showing that it could not have been the case that Tarski still had a universalistic view of logic in *Ps*, then by arguing that Tarski abandoned type theory.

A rough sketch of de Rouilhan's argument against a universalistic standpoint in *Ps* can be presented as follows. Suppose that there is a universal language, say U. Observe then that there is no conclusion in the Postscript analogous to conclusion B in *Wb*.[14] There is, then, no conclusion that for certain languages the definition of truth is impossible. Indeed from the text in *Ps* we can even conclude the opposite, i.e. that it is always possible to construct a language of a higher order than any given language, [15] and thus it follows from conclusion A* that it is always possible to construct a definition of truth for sentences of the latter language in the former. Now take the supposed universal language U. *Ps* leads us to believe that there is a language of a higher order than U, which is in contradiction with the universality of this language. Hence we must conclude that there is no universal language for Tarski in *Ps*.

Field's interpretation, as discussed in the previous section, offers a way out of this conclusion. It may not be Tarski's view—according to this interpretation—that it is absolutely *always* possible to construct a definition of truth for every language, but only for those languages which Tarski was implicitly considering: languages in which the order of the variables is bounded by an ordinal, be it finite or infinite. The role of universal language could then still be played by a language of which the order of variables is unbounded by any ordinal.

For the moment, however, let us accept de Rouilhan's argumentation in order to continue the story. The second step of de Rouilhan's argumentation is to show that Tarski abandoned type theory. De Rouilhan suggests that Tarski's retraction of the principles of semantical categories in *Ps* signals primarily his abandonment of the principles of the theory of types (rather than those of the theories of Leśniewski or Husserl) (de Rouilhan 1998, p. 93). Furthermore, it is suggested that "variables of indefinite order" have no place in Russell's theory. This latter reason is argued for as follows:

[14] De Rouilhan, like Field, holds that a conclusion analogous to conclusion *C* of *Wb* can easily be provided for *Ps*. Yet contrary to Field's proposal, de Rouilhan proposes that this conclusion runs somewhat as follows (de Rouilhan 1998, p. 98):

> $\overline{C}$ *On the other hand, even if the order of the metalanguage is at most equal to that of the language itself, the consistent and correct use of the concept of truth is rendered possible by including this concept in the system of primitive concepts of the metalanguage and determining its fundamental properties by means of the axiomatic method.*

[15] In particular it is always possible to construct the metalanguage in such a way that it contains variables of higher order than all the variables of the language studied. The metalanguage then becomes a language of higher order and thus one which is essentially richer in grammatical forms than the language we are investigating. (Tarski 1956, pp. 271–272)

> From the perspective of the latter theory [set theory], the variables that represent names of sets are the same (and are thus of the same category and the same order syntacticly speaking) as those that represent the names of possible elements of these sets, etc., and, finally, as those that represent the names of individuals. It is precisely in this way that the objects of higher order in the ontological sense fully deserve the name of “sets”. From the standpoint of simple type theory, however, the difference in the ontological sense of order implies a difference in the syntactic sense of order. The variable representing the names of higher-order objects in the ontological sense are themselves higher order in the syntactic sense.[16] (de Rouilhan 1998, p. 92)

It is exactly these ideas which I will question further on. I will argue that in Russell’s type theory the connection between syntactic higher order and ontological higher order is not so clear-cut, and with that I will show that the principles of semantical categories are not so essential as de Rouilhan suggests. This is not to say that Tarski wasn’t slowly moving towards an anti-universalistic, set-theoretical stance, but rather that *Ps* does not in fact evince a radical change in his attitude toward logic.

To summarise, in de Rouilhan (1998) two notions of order[17] are explained: a syntactic notion and an ontological notion. For Zermelo’s set theory the ontological notion is the most fruitful one. It is with respect to the ontological notion that de Rouilhan distinguishes variables of indefinite order in a set theoretical setting.

For type theory, on the other hand, the syntactical notion and the ontological notion are closely linked, de Rouilhan argues, in such a way as to leave no room for variables of indefinite order. Together with the idea that type theory is naturally seen as universalistic and the conclusion of *Ps* is anti-universalistic, this leads de Rouilhan to conclude that *Ps* cannot be interpreted in a type theoretical setting.

5 Towards transfinite type theory

In the previous section we have seen that de Rouilhan emphasises the transition from a universalistic point of view in *Wb* to an anti-universalistic point of view in *Ps*. Moreover, this transition was accompanied, in his view, by a change from a type theoretical framework to some other framework, possibly that of set theory. Although the anti-universalism claim can be countered by Field’s interpretation of *Ps*, this cannot by itself dispel de Rouilhan’s claim that the framework has changed, for this latter claim is strengthened by the idea that the syntactical order mirrors the ontological order of

[16] Du point de vue de cette dernière théorie, les variables qui représentent des noms d’ensembles sont les mêmes (et sont donc de même categorie et de même ordre au sense syntaxique) que celle qui représentent des noms d’éléments éventuels de ces ensembles, etc., et, finalement, que celles qui représentent des noms d’individus. C’est dans cette mesure précisément que les objets d’ordre supérieur au sens ontologique méritent pleinement le nom d’“ensembles”. Du point de vue de la théorie de types simple, au contraire, la difference d’ordre au sens ontologique implique une différence d’ordre au sens syntaxique. Les variable qui représentent des noms d’objets d’ordre supérieur au sense ontologique sont elle-même d’ordre supérieur au sens syntaxique.

[17] In fact, de Rouilhan distinguishes *three* notions of order: the syntactical and the ontological which are orders given to languages, and additionally an order given to theories.

objects in the theory of types. This means in particular that we would not be able to fit variables of indefinite order in simple type theory.

At the same time de Rouilhan does not answer the crucial question of what (kind of) formal languages Tarski had in mind in *Ps*, if not the language of set theory. Recall that Tarski says that *it is only one step* from the considered languages to the language of set theory, not that the considered language *is* the language of set theory. He also makes explicit that this step requires a change in the concept of order.

In this section I will argue, *pace* de Rouilhan, that Tarski may actually still have been working in a kind of type theoretical framework, *like* simple type theory, possibly the theory of transfinite types. Such a generalisation can be found in the interpretation of Russell's type theory in Carnap's theory of levels.[18] The theory of transfinite types is not the same as set theory, although by lifting only one restriction from the theory that Tarski seems to describe in *Ps* we can pave the way for obtaining a natural translation to set theory.

In order to identify variables of indefinite order in transfinite type theory, it is informative to consider the origins of this theory, and especially its relation to "typical ambiguity" in simple type theory.

5.1 Origin of variables of indefinite order

To show where the variables of indefinite order come from, let us go back to the roots of type theory. In 1908 Russell published his own solution to the paradoxes of logic, namely his ramified theory of types (Russell 1908; see also Whitehead and Russell 1910). Characteristic of the ramified theory of types are the two hierarchies on which it is based (see also Feferman 2002): the hierarchy of *types*, and the hierarchy of *orders*. Types in this context are not the same as Tarski's types (see Sect. 2) and not the same as semantical categories. For Tarski, semantical categories (and types) primarily concern *expressions* (so, it is a *syntactic* notion), whereas for Russell they primarily concern *objects* (an *ontological* notion). Although the details of the ramified theory of types do not concern us here, we will go into it just a little further to explain its relation to the simple theory of types of concern not only to de Rouilhan, but also to Tarski:

> From the formal point of view the theory of semantical categories is rather remote from the original theory of types of (Whitehead and Russell, 1910, pp. 37 ff.); it differs less from the simplified theory of types (cf. Chwistek (1924), pp. 12–14; Carnap (1929), pp. 19–22) and is an extension of the latter. (Tarski 1956, fn 1, p. 215)

The simple theory of types omits the hierarchy of orders of the ramified theory.

There are two ways in which Russell's type theories appear flexible with respect to type allocation. First, in his early work (1905–1907), some variables range over everything in his ontology, and so are not restricted to any certain type (Proops 2007,

[18] Such a generalisation as that to which we are referring is also alluded to in Hilbert (1926) and Gödel (1931), and is thus not *distinctive* for the theory of levels.

pp. 7 and 24). In Russell (1908) we can still find some remnants of this idea in the fact that so-called "real" variables are not typed.

In Russell (1908) Russell, like Tarski, also located the problems, such as the Liar paradox, in the fact that certain propositions are self-referential. However, where Tarski solved the issue on the level of languages—clearly distinguishing an object language and a meta-language—Russell's solution can be said to be on the level of propositions within one and the same language.[19]

One of the main ingredients of Russell's solution is the separation of the notions "any" and "all". Whereas he admitted "all" (universal quantification) in the construction of a new proposition, a construction with "any" leads us outside the set of propositions and thus enables us to speak *about* a proposition without *using* a proposition. Going back to the paradoxes, we see that this method may thus solve the problem of self-referring propositions by only allowing other "things", outside the totality of propositions (whatever this may mean), to speak about all propositions.

Here is an example. Russell addresses the Law of Thought: "All propositions are either true or false." Because it is a proposition itself, constructed using the universal quantifier "all", it falls within its own scope, which is what Russell wants to avoid (Russell 1908, p. 226). Instead, therefore, Russell reformulates this law as follows: "*p* is true or false, where *p* is *any* proposition" (Russell 1908, pp. 229–230). This latter statement falls outside the totality of propositions and does thus not speak about itself. Russell concludes:

> Thus we may admit "any value" of a variable in cases where "all values" would lead to reflexive fallacies; (Russell 1908, p. 230)

Real variables are the ones bound by "any" and apparent variables are bound by "all". Because it is apparent variables that occur in propositions, it is these that call for careful treatment so as not to reintroduce logical paradoxes. For this reason Russell introduces *types*, which guard the correct use of apparent (not "real"!) variables:

> Whenever an apparent variable occurs in a proposition, the range of values of the apparent variable is a type, the type being fixed by the function of which "all values" are concerned. (Russell 1908, pp. 236–237)

One could argue that the untyped variables of Russell's type theory are of *indefinite order*.[20,21]

A consequence of the above example of the Law of Thought also illustrates the second way in which type allocation may be thought of as flexible in the practice of type theory. We can, in fact, conclude from that example that there cannot be a totality

[19] Russell's and Tarski's solution differ on other points as well, such as on the intentionality/ extensionality issue. See Church (1976, p. 756).

[20] It may be exactly because the real variables are not restricted to a certain type that Russell did *not* make his types visible in the syntax of the ramified theory of types (see Church 1976, fn 9, p. 750).

[21] Intuitively, given our modern understanding of logic and logical notions, one could say that apparent variables are bound whereas real variables are free (see e.g. Church 1976, p. 750). This identification seems, however, not only slightly inaccurate (the real variables are in fact bound, to wit by "any"), but also neglects a very important difference between apparent and real variables: the former are typed, whereas the latter are not.

of propositions. If one takes any set of propositions, say S, then the statement "All propositions of S are either true or false" is itself a proposition. And it has to be a proposition not included in S, because a proposition is not allowed to fall into its own scope (see Whitehead and Russell 1910, p. 39).

The fact that there is no totality of propositions in type theory leads, however, to the following problem: What is the domain of the predicate "x is false" (Whitehead and Russell 1910, p. 44)? It cannot be the totality of propositions, but the predicate still seems to make sense for any proposition that we substitute for x. Because the type of "x is false" is based on the types of the arguments it can take, it seems that we cannot assign a type to it.

Russell's solution to this problem is to say that, although indeed we cannot assign a single type to this predicate, we can assign a type to it in each context in which it appears. There is thus not one predicate "x is false", but there are many, each of a different type, depending on the kind of propositions to which it is applied. So it only appears that any proposition might appear as argument, but that is due to *systematic* or *typical ambiguity* (Whitehead and Russell 1910, p. 45): "x is false" can denote many different predicates, although they all look the same (we leave the type ambiguous).

Similarly, in a fully explicitly typed language the inclusion sign ("$\subset$"), for example, should have a fixed type. More precisely put: the type should be one higher than that of both of its arguments. This also entails that the language should contain a separate inclusion sign for each type. In order to overcome this cumbersome practice Russell held signs like "$\subset$" (and both arguments, if variables) to be typically ambiguous. That is to say: no fixed type is allocated to the sign, but any specific occurrence of the sign could be given a certain type (in a systematic way; see Feferman 2004, p. 135 and Quine 1938, p. 130).

One may argue that this practice is not really an example of the flexibility of type theory, but rather a flexible way to handle type theory. By turning this pragmatic flexibility into a formal property of the system itself, transfinite type theory can be obtained.

5.2 From simple type theory to transfinite type theory

From typical ambiguity it is only a small step to a transfinite type theory. In the case of "$\subset$", for example, instead of allowing it to be systematic ambiguous in one's language as a kind of innocent shorthand, one could from the very beginning admit it as a sign that can have arguments of any (finite) type as a formal part of the language. These signs, which can have arguments of any (finite) type, can then themselves also be typed and they belong to order ω.[22]

Carnap introduced "levels", which are very similar to Tarski's orders in *Ps* (see Coffa 1987). Carnap's levels, like Tarski's orders, can be numbered by either a finite or a transfinite ordinal number. Furthermore, the level of the arguments of a predicate is not fixed once and for all: say, the nth argument place of some predicate may be occupied by an argument of level i in one context, and by an argument of level $j\,(\neq i)$

[22] Note that giving type ω to the predicate "x is false" would not solve anything, though.

in another (Carnap 1934, p. 188). Carnap makes the following remark concerning a transfinite levels and systematic ambiguity (though similar constructions can be, and have been made outside of Carnap's framework of levels):

> In [*Princ. Math.*] Russell has used the symbol '$\subset$' and many others with arguments of any (finite) level whatever, so that, according to our definition, they belong to the level ω. Russell does not, however, attribute a transfinite level to these, but interprets their mode of use as "systematic ambiguity". (Carnap 1934, p. 189)

Another example of a symbol with level ω would then be "=" (identity), for it can also hold arguments of any (finite) level.

Not only does this show where in Russell's type theory the expressions of transfinite order come in; we can also see how their use leads easily (again) to variables of indefinite order. Suppose that one wants to express that everything is equal to itself. Instead of expressing this for each type separately, using a variable α of indefinite order this can be expressed as "$\forall\alpha.\alpha = \alpha$".

So we see that we can have it all—variables of indefinite order, as well as expressions of transfinite order—without making the big leap to set theory. It is not the case that in simple type theory, or in its extension into a transfinite type theory, a difference in the ontological sense of order implies a difference in the syntactic sense, or at least not so strictly as de Rouilhan seems to assume.

Nor is it the case that by allowing transfinite types, one has moved into set theory. Some set theory is necessary in order to formulate the theory of transfinite types, but that is a different matter altogether than working *in* set theory.

5.3 From transfinite type theory to set theory

To get a straightforward translation from simple type theory to set theory, more is needed than merely to allow transfinite types and variables of indefinite order, though not that much more. Gödel distinguishes three "superfluous restrictions" that have to be removed in order to obtain ZF set theory from simple type theory (Gödel 1933). As explained in Linnebo and Rayo (2012, p. 273), removing these restrictions amounts to:

1. Allowing infinite types;
2. Allowing the type structure to be cumulative;
3. Allowing a type-unrestricted notion of predication.

One then obtains straightforward translations from type theory into set theory and vice versa.

The first step we have already discussed above. The second step refers to the fact that a class in (finite) simple type theory can only contain classes of one and the same type among its elements, whereas the members of a set can be diverse in that respect. If one moves to transfinite type theory, this pureness of types causes a problem: For what is the type of the arguments to which an expression of type ω applies? It seems natural to define that such an expression applies to arguments of any finite type, and, as we have seen, that is also what Carnap did. This means that type ω becomes *cumulative*, and, with a similar argumentation, so does any other limit ordinal. By also allowing successor types to be cumulative, we finish the second step.

The third step refers to predications that are not well-formed according to the formation rules of type theory, because the argument is of a higher type than the predicate. Instead of dismissing such a predication as ill-formed or meaningless, it should be allowed into the language, but considered to be false.

It can now easily be seen that Tarski proposed to take the first two steps in *Ps*, but not the third. Thus it is the case not only that Tarski is not working in set theory, but also that a straightforward translation to set theory as explicated in Linnebo and Rayo (2012) is unavailable.

6 Tarski's stance towards type theory

Additional evidence concerning Tarski's stance on universalism, type theory and set theory in *Ps* may be obtained through many of his later writings (see e.g. Rodríguez-Consuegra 2005).[23] However, I will mention here only Tarski (1944), because its content relates directly to Tarski (1935): it explains and summarises the results of the latter paper.

In Tarski (1944) the notion of a language being *essentially richer* than another one is used in the explanation of these results. Using that notion, the conclusions of *Ps* can be reformulated as follows: the meta-language is essentially richer than the object language if and only if the definition of *true sentence* of the object-language can be constructed in the meta-language. Tarski writes:

> It is not easy to give a general and precise definition of "essential richness." If we restrict ourselves to languages based on the logical theory of types, the condition for the meta-language to be "essentially richer" than the meta-language is that it contain variables of higher logical type than those of the object-language. (Tarski 1944, p. 675)

This shows that although Tarski may not have been thinking exclusively about languages based on the theory of types, he does not exclude them, for whatever reason, from the conclusion of *Ps*. His explanation of essential richness given in the passage cited above for languages based on the theory of types seems even to match well the conclusions of *Ps*: we have only to replace "type" by "order."

Moreover, the fact that Tarski explained the results of his investigations into the notion of *true sentence* in terms of type theory is in full agreement with the observation that "the logical framework of Tarski's investigations in the Thirties was type-theoretic" (Bellotti 2003, p. 409; see also Betti 2008, p. 55).

[23] Especially relevant is Tarski and Corcoran (1986). In that paper Tarski compares type theory and first-order set theory with respect to the question whether or not the membership relation is a logical notion. At least it shows that Tarski had not completely abandoned research to type theory in 1966.

7 Conclusion

The anti-universalistic stance has been defended by interpreting the absence in the Postscript of a conclusion which states the undefinability of the "true sentence" for certain languages as a denial of such a conclusion. If we take instead the more cautious view that nothing definite can be concluded from this absence—as Field does—the argument in favor of Tarski's anti-universalism ceases to be convincing.

One of the consequences of Tarski's supposed anti-universalism would be his move away from type theory. Countering the argument in favor of Tarski's anti-universalism wouldn't suffice also to undermine the idea of Tarski's move away from type theory, because there are other, more syntactical issues that also seem to gesture towards this move. We have shown, however, that type theory is flexible enough to accommodate these syntactical issues.

This means that we have found no evidence for a radical change in the Postscript, either in its logical framework or in the philosophical ideas it expresses, as compared to the main text of the *Wahrheitsbegriff*. Or, at least: we have found evidence neither of a change from universalism to anti-universalism, nor of a clear-cut change from syntactical order to ontological order, nor of a change from type theory to set theory.

If we should characterise the Postscript in terms of change at all, it would be in another sense: we can see it as the final desertion of Leśniewski's systems in favor of Russell's type theory. We have seen that the theory of semantical categories is not essential for type theory and that its abandonment thus cannot be understood as a desertion of the latter theory. On the other hand, the theory of semantical categories *is* essential to Leśniewski's systems. So, whereas the main text of the *Wahrheitsbegriff* already flirted abundantly with type theory, the Postscript, with its rejection of the theory of semantical categories, can only be a sign that the influence of Leśniewski's work on Tarski had by this point run its course.

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